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(54) **FLAMMABLE GAS DILUTION**

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None

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

3,953,002 A 4/1976 England, Jr. et al.

6,257,754 B1 7/2001 SØndergaard

(Continued)

FOREIGN PATENT DOCUMENTS

CN 1405443 A 3/2003

CN 1234463 C 1/2006

(Continued)

OTHER PUBLICATIONS

Japanese Notice of Reason for Rejected dated Dec. 24, 2024 for corresponding Japanese application Serial No. 2022-562516, 4 pages.

(Continued)

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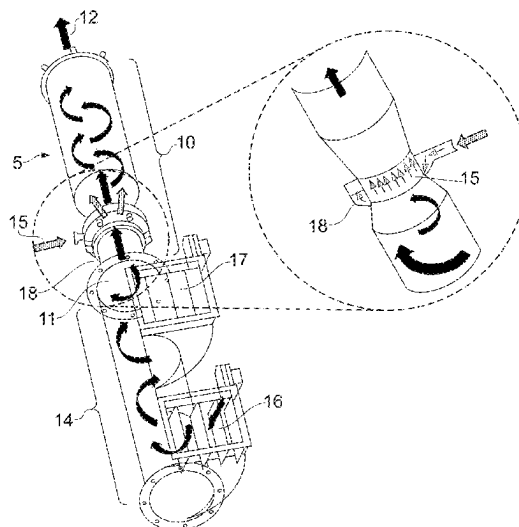
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(57)

ABSTRACT

A flammable gas diluter includes: a dilution vessel comprising an outer envelope defining a longitudinal flow passage from an inlet to an outlet; at least one air inlet for directing a flow of air into the inlet of the diluter; and a flammable gas inlet arrangement. The dilution vessel has a plurality of gas flow directing formations arranged between the flammable gas inlet arrangement and the outlet, each being at a different position along a length of the dilution vessel. At least one of the plurality of gas flow formations is an inwardly directing gas flow formation for directing gas flow away from the outer envelope and at least one of the gas flow formations is an outwardly directing gas flow formation for directing gas flow towards the outer envelope.

15 Claims, 6 Drawing Sheets



(51)	Int. Cl.		CN	203490552 U	3/2014
	B01F 25/313	(2022.01)	CN	104696611 A	6/2015
	B01F 25/431	(2022.01)	CN	105435669 A	3/2016
	B01F 35/21	(2022.01)	CN	105556211 A	5/2016
	B01F 35/22	(2022.01)	CN	106237880 A	12/2016
	B01F 35/221	(2022.01)	EP	1023932 A1	8/2000
	B01F 35/71	(2022.01)	GB	675319 A	7/1952
	G03F 7/00	(2006.01)	GB	2561899 A	10/2018
			JP	S54-59569	5/1979
(52)	U.S. Cl.		JP	H64-3813	3/1989
	CPC	B01F 25/3131 (2022.01); B01F 25/4312	JP	2001227783 A	8/2001
		(2022.01); B01F 25/431952 (2022.01); B01F	JP	2003003910 A	1/2003
		35/2132 (2022.01); B01F 35/2202 (2022.01);	JP	2003117367 A	4/2003
		B01F 35/2211 (2022.01); B01F 35/7181	JP	2009136716 A	6/2009
		(2022.01); G03F 7/70841 (2013.01)	TW	200614384 A	5/2006
			TW	200738322 A	10/2007
			WO	2008108812 A2	9/2008
			WO	2011128824 A1	10/2011

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,650,973 B2 *	2/2014	Wei	G01N 1/2252	73/863.03
2004/0251566 A1	12/2004	Kozyuk		
2006/0070675 A1	4/2006	Hsu		
2006/0191254 A1	8/2006	Bui et al.		
2008/0273995 A1 *	11/2008	Bailey	F04D 27/0292	417/279
2012/0014209 A1	1/2012	Smith		
2020/0047225 A1 *	2/2020	Bailey	F04D 27/0292	
2021/0106955 A1	4/2021	Matsushima et al.		
2021/0291127 A1 *	9/2021	Hishikawa	F23N 1/025	

FOREIGN PATENT DOCUMENTS

CN	101808721 A	8/2010
CN	102084182 A	6/2011
CN	103111212 A	5/2013

OTHER PUBLICATIONS

Taiwanese Search Report dated Oct. 11, 2024 and Office Action for corresponding Taiwanese application Serial No. 110113696, 8 pages.
 British Search Report dated Oct. 15, 2020 for corresponding British Application No. GB2005545.5, 1 pages.
 PCT Notification of Transmittal of the International Preliminary Report on Patentability dated Jun. 24, 2021 and PCT International Search Report and PCT Written Opinion dated Jun. 24, 2021 for corresponding PCT application Serial No. PCT/GB2021/050852, 12 pages.
 First Chinese Office Action dated Sep. 29, 2023 and Search Report dated Sep. 25, 2023 for corresponding Chinese Application No. CN202180028602.6, 23 pages.
 Chen Shuyi et al., "Fuel Combustion and Combustion Device," Metallurgical Industry Press, p. 366, Figs.7-18 and p. 367, para.1, dated May 31, 1985.

* cited by examiner

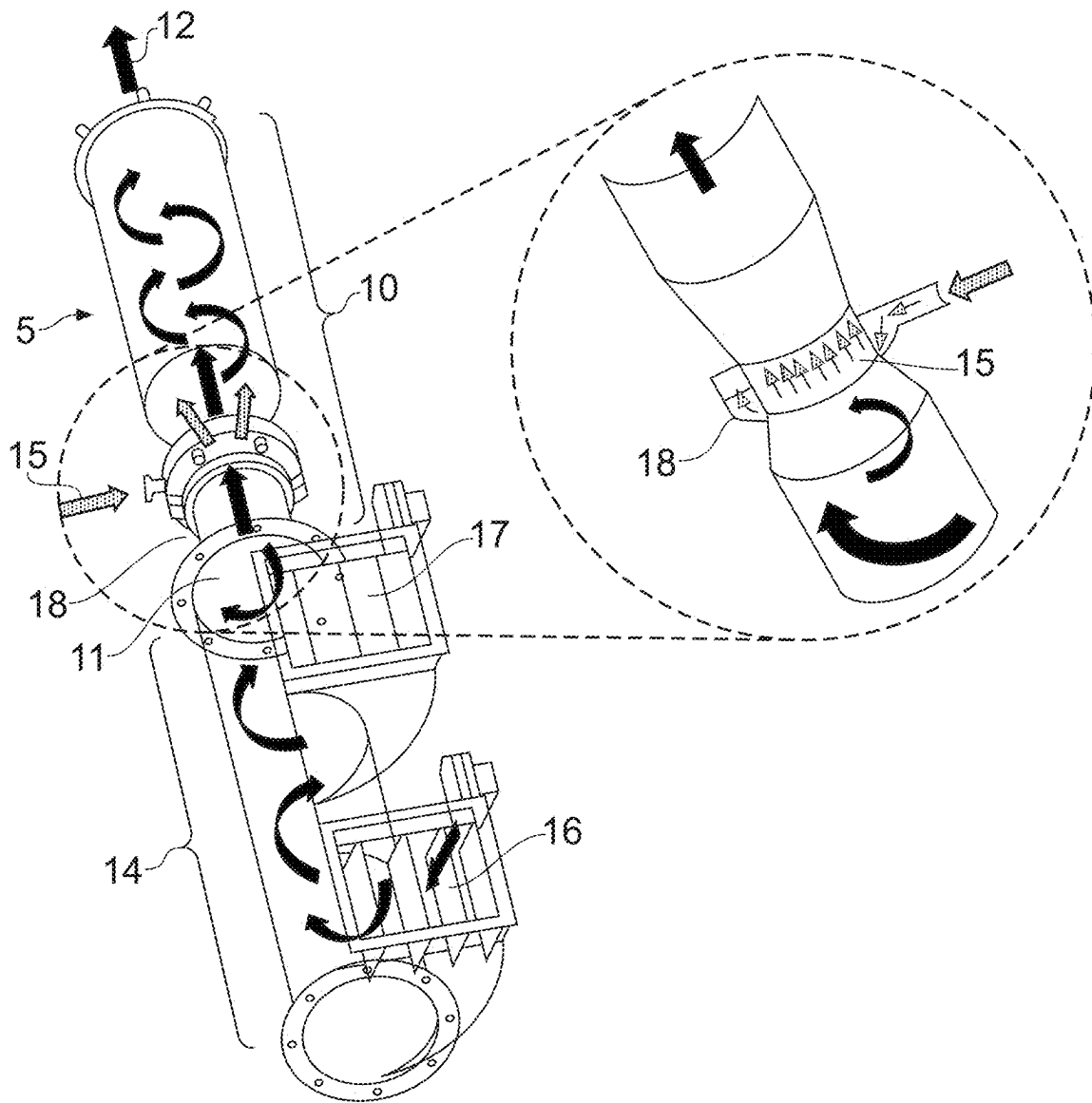


FIG. 1

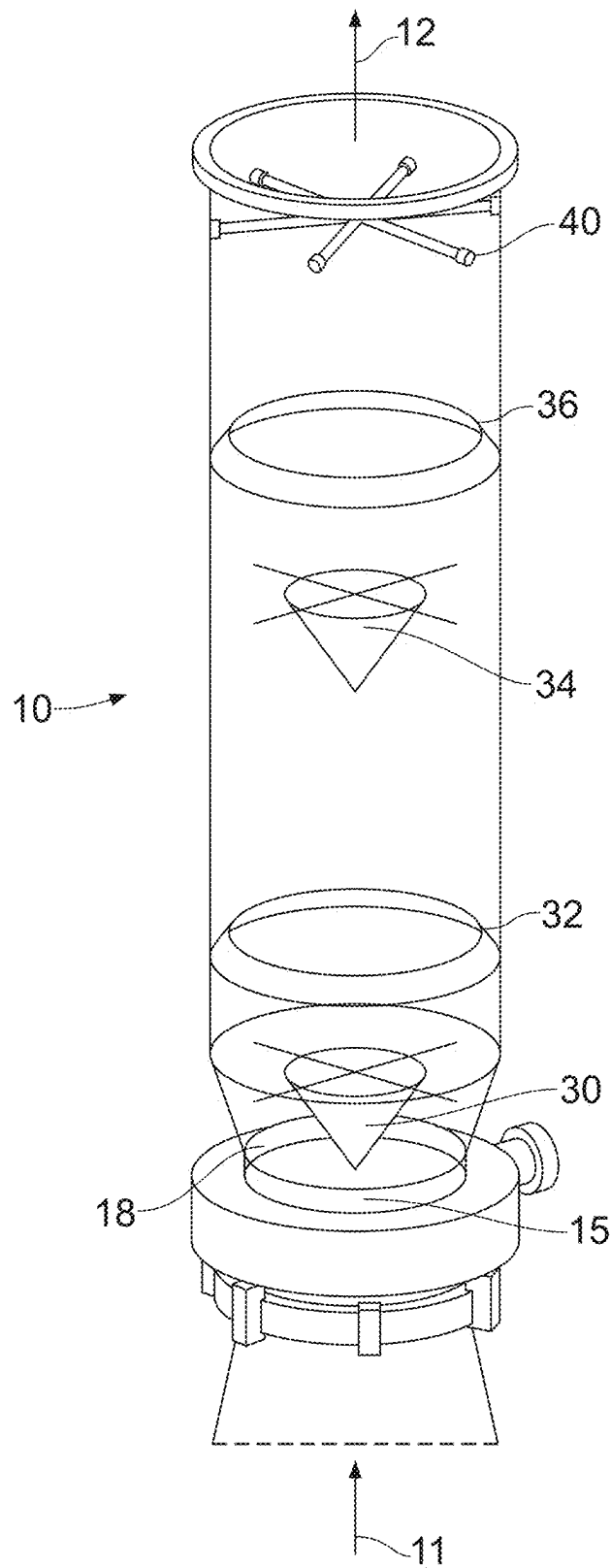


FIG. 2

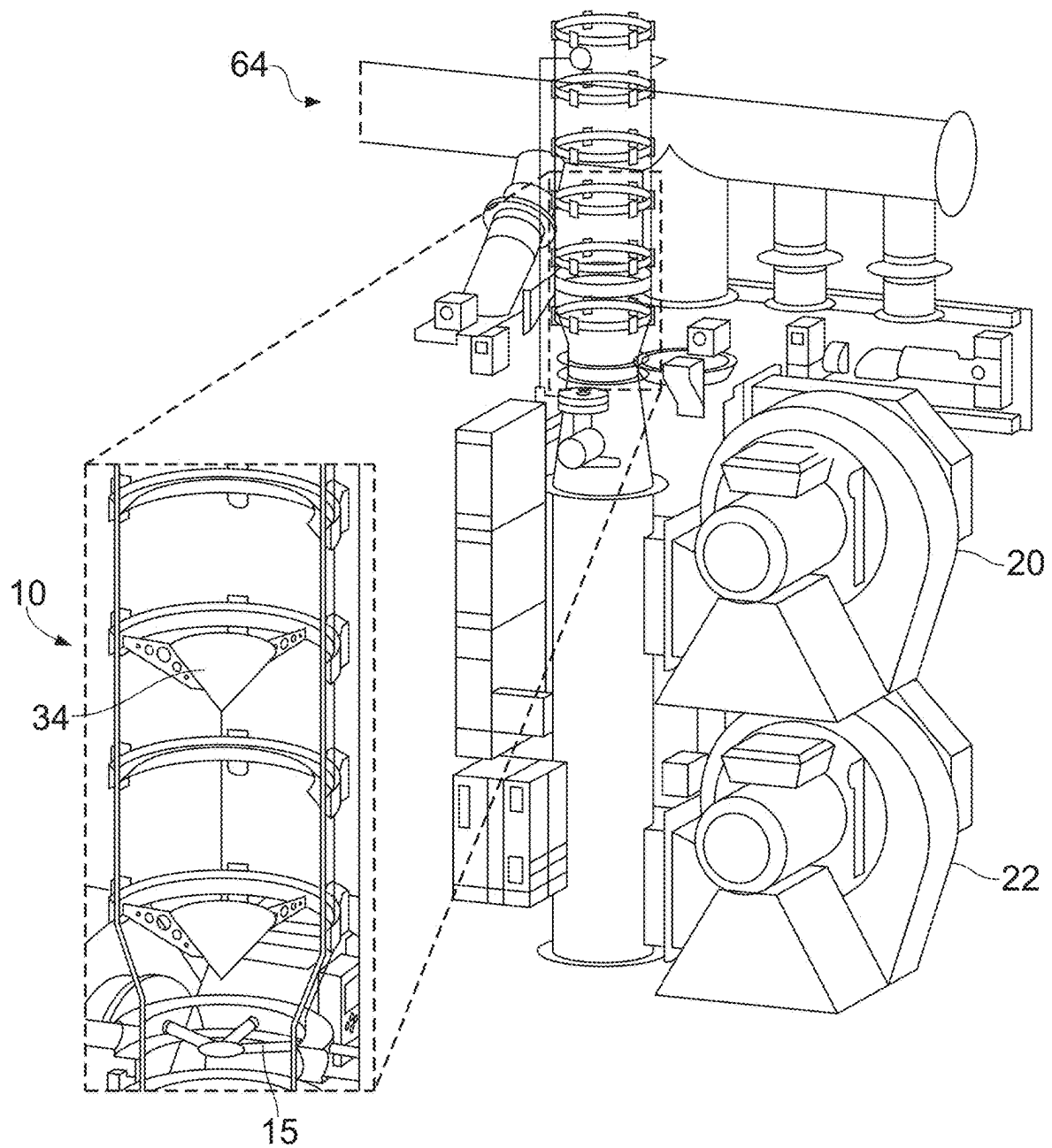
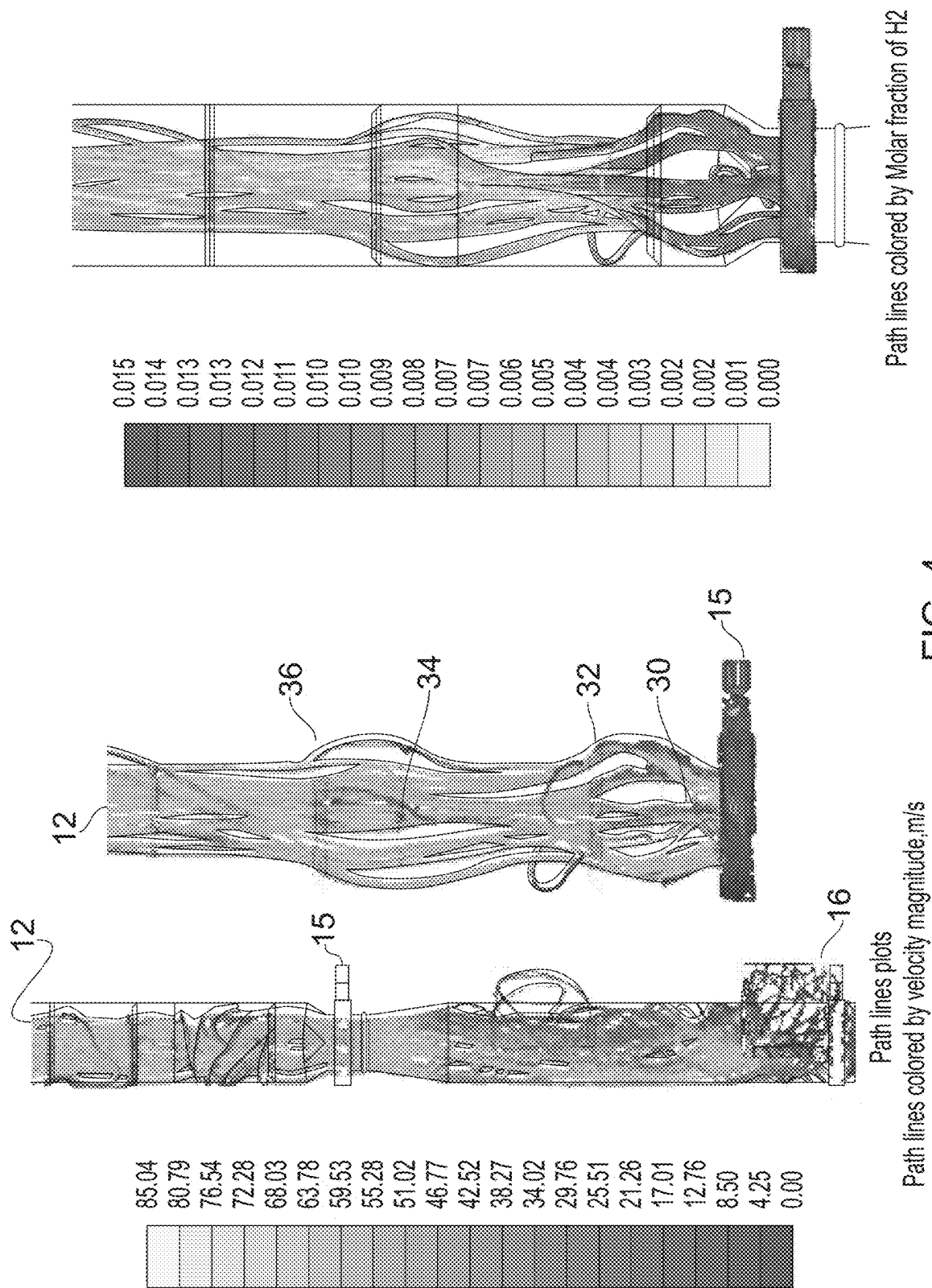


FIG. 3



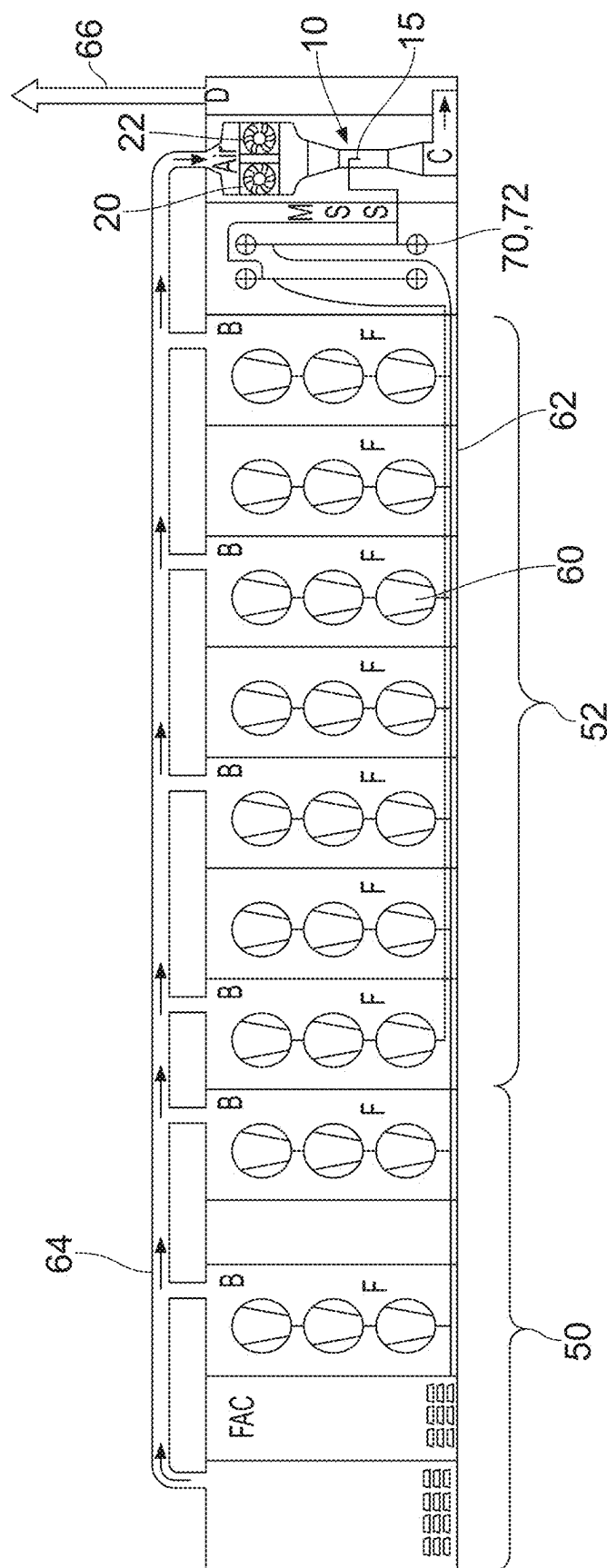


FIG. 5

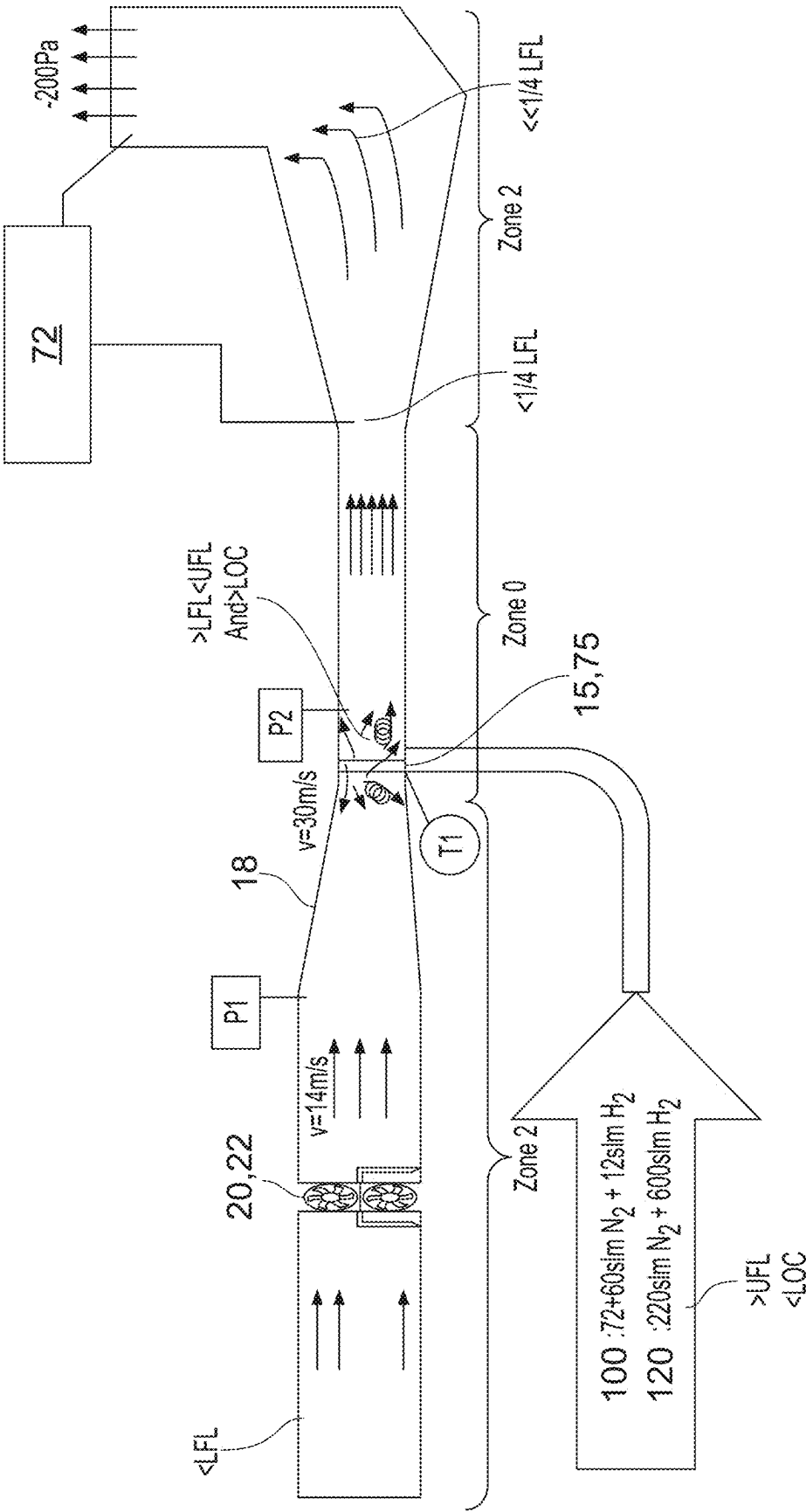


FIG. 6

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FLAMMABLE GAS DILUTION**CROSS-REFERENCE OF RELATED APPLICATION**

This application is a Section 371 National Stage Application of International Application No. PCT/GB2021/050852, filed Apr. 7, 2021, and published as WO 2021/209742A1 on Oct. 21, 2021, the content of which is hereby incorporated by reference in its entirety and which claims priority of British Application No. 2005545.5, filed Apr. 16, 2020.

FIELD

The field of the invention relates to flammable gas dilution and in some embodiments to a vacuum pumping and abatement system.

BACKGROUND

There are semiconductor fabrication processes where the gas to be exhausted and abated is a flammable gas such as hydrogen. In lithography for example, the product is manufactured by controlled exposure to a source of radiation. In this case the source of radiation is extreme ultraviolet EUV radiation. In this process hydrogen is used as a curtain gas in increasingly large quantities to shield the optics and mirrors from the sputtered tin that is excited by the laser to radiate EUV light in the lithography tool. These processes are performed within a vacuum and the vacuum system provides the vacuum pressures necessary for this process to occur and carries the hydrogen away to be safely abated.

In many abatement systems the flammable gas that is removed from the vacuum process chamber is burned to remove the gas. There are environmental implications associated with this and it generally requires two abatement tools, an operational one and a backup one in case the burner in the operational abatement tool is extinguished. This arrangement is expensive in both fuel, and space.

It would be desirable to provide an alternative way of abating flammable gas from a gas stream, so that the gas stream can be safely exhausted.

The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the background.

SUMMARY

A first aspect provides a flammable gas diluter for diluting a flow of flammable gas to a concentration below the flammability limit of said flammable gas, said diluter comprising: a dilution vessel comprising an outer envelope defining a longitudinal flow passage from an inlet to an outlet; at least one air inlet assembly for directing a flow of air into said inlet of said dilution vessel; a flammable gas inlet arrangement comprising a plurality of apertures at least some being arranged at different distances from said outer envelope of said dilution vessel across a cross section of said dilution vessel towards said inlet end of said dilution vessel; a plurality of gas flow directing formations arranged between said flammable gas inlet arrangement and said outlet, each being at a different position along a length of said dilution vessel; wherein at least one of said plurality of gas flow formations is an inwardly directing gas flow

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formation for directing gas flow away from said outer envelope and at least one of said gas flow formations is an outwardly directing gas flow formation for directing gas flow towards said outer envelope.

Flammable gas is challenging to handle and dispose of. For these reasons flammable gas output by a system has generally been dealt with by burning it in a burner, sometimes using methane as a fuel. This has implications for the environment, is relatively expensive in fuel and space and can have reliability issues. Despite the challenges associated with handling flammable gas, were one able to safely dilute it to below its flammability limit then many flammable gases could simply be vented to atmosphere.

Previously where flammable gases have been diluted this has generally been done with nitrogen which is an inert gas. However, in systems where the amount of flammable gas is high, the amount of nitrogen that would be required to reduce the concentration of the flammable gas to below the flammable gas flammability level is in many cases prohibitively expensive.

The dilution of flammable gas would seem to be an acceptable alternative to burning flammable gas as a means of abatement provided that one could do so safely. The inventors of the present invention recognised that many of the hazards associated with handling flammable gases are associated with transporting the flammable gas from the point of use to a point where it can be safely abated. Thus, many of the challenges can be addressed by providing a diluter that is simple in design, relatively compact and has few moving parts, allowing it to be used as a point of use diluter to dilute flammable gas at a point at or close to where it has been used.

A further advantage of a diluter according to one aspect of the invention is that owing to the simple design and the paucity of moving parts, the diluter is both robust, reliable and unlikely to fail. This can be very important in a system where the failure of an abatement unit may cause the flammable gas to shut off to the system which in turn may require the system to be shut down immediately which can result in damage to the system.

Said dilution vessel comprises a constricted portion, said flammable gas inlet arrangement being located within said constricted portion such that said air is accelerated prior to passing said flammable gas inlet.

When the flammable gas is input to the diluter it may initially be above its upper flammability limit, but on dilution it will become flammable until it is diluted to below its lower flammability limit and thus, steps need to be taken to mitigate any risk of ignition between these points. Increasing the air flow at the flammable gas inlet by using a constriction, is one way of diluting the gas relatively quickly initially and inhibiting ignition particularly if the increased flow rate is above the flame speed for the flammable gas.

The constriction may only extend for a fraction of the length of the dilution vessel, the dilution vessel expanding out beyond the constricted portion such that the gas flow slows. This helps mixing between the flammable gas and air within a constrained space.

In some embodiments, the flammable gas diluter comprises at least one gas flow generator for pumping a flow of air into said air inlet assembly of said flammable gas diluter, said at least one gas flow generator being located upstream of said flammable gas inlet arrangement.

As noted above the concentration of flammable gas within the diluter is between the upper and lower flammability limits for a portion of the length of the diluter downstream of the flammable gas inlet. Thus, it is advantageous for any

potential sources of ignition to be absent for this portion of the diluter and the diluter is configured such that any moving parts such as exist in a gas flow generator, which may be in the form of a fan, are upstream of the flammable gas inlet, so that there are no moving parts which might be potential causes of ignition downstream of the flammable gas inlet.

In some embodiments, the flammable gas diluter comprises two gas flow generators configured to operate as operational and backup gas flow generators.

In many systems where flammable gasses need to be removed and abated it is important that the concentration of flammable gas does not rise above a certain level and thus, it is important particularly where the flammable gas is being diluted at the point of use that the system is reliable and does not fail. Thus, in some embodiments two gas flow generators are provided, one configured to operate as the primary gas flow generator and the other as a backup gas flow generator that functions were the primary gas flow generator to have a fault. As the diluter is a simple design with few moving parts, a reliable system can be provided where the components that do have moving parts, in this case the gas flow generators are duplicated.

In some embodiments, said air inlet assembly comprises a conduit directing air to said inlet of said dilution vessel, said two gas flow generators being located at substantially a same longitudinal position on said conduit, in some embodiments at different positions around a perimeter of said gas conduit.

In some embodiments the two gas flow generators may be located at different longitudinal positions along the conduit in a vertical arrangement, one on top of the other. In other embodiments they may be located on either side of the conduit which reduces the height of the diluter and makes it more compact.

In some embodiments, said at least one gas flow generator and dilution vessel are configured such that a speed of flow of said air at said flammable gas inlet arrangement is greater than a flame speed of flammable gas.

As noted previously, it is important to impede ignition of the flammable gas when it is within the diluter particularly close to the inlet of the flammable gas where the concentrations of flammable gas will be highest. Thus, providing the air at the flammable gas inlet arrangement at a speed that is greater than a flame speed of the flammable gas will impede its ignition. In this regard, the flame speed of hydrogen for example is between 3 to 4 meters per second such that a hydrogen diluter should be configured so that the air speed at the hydrogen gas inlet is greater than 3 to 4 meters per second. In some embodiments it is configured to be greater than 20 meters per second, preferably greater than 25 or 30 meters per second.

In some embodiments, in addition to providing this accelerated flow of air there may also be a flame arrester cone at the flammable gas inlet.

In some embodiments, said flammable gas inlet arrangement is configured such that said apertures are facing away from said gas outlet.

It has been found that improved mixing of the flammable gas with the air occurs where the apertures are facing away from the gas outlet and in some embodiments facing towards the gas inlet.

In some embodiments, said apertures of said flammable gas inlet arrangement are between 2 and 5 mm in diameter.

The choice of size of a flammable gas apertures affects the flow of flammable gas into the diluter, this is particularly so where the apertures are facing towards the diluting air flow. Too large an aperture and the air flow will impede exit of the

flammable gas and may indeed cause contamination with air flowing into the vacuum system while too small an aperture will inhibit the gas flow. An aperture size of between 2 and 5 mm diameter has been found to provide particularly effective flow of flammable gas into the diluter.

In some embodiments, said flammable gas inlet arrangement comprises an outer ring channel and radial channels running from said outer ring channel towards a centre of said ring, said radial channels comprising said apertures.

In order to provide the flow of flammable gas into the air flow in a way that encourages mixing between the two gas flows, it has been found to be advantageous to provide the apertures in the inlet arrangement at different radial positions across the gas flow such that there is not a single plume that is formed but rather several flows of flammable gas from different apertures at different radial positions. The use of arms across the cross section of the diluter has also been found to provide an effective arrangement where the air flow is not unduly inhibited and thus, the vacuum pumps are not over pressured. The collar or outer ring channel surrounding the diffuser where the hydrogen is introduced from the pumping system into the diffuser, is sized to permit substantially equal and unrestricted flow to the spider arms around the inner circumference. In some embodiments, the spider comprises between 4 and 8 arms across the diffuser or diluter inlet. The spider arms are sized so as not to restrict the diffuser air flow path by more than 30%. The apertures in the spider arms are distributed substantially evenly along the length of the arms and arranged to face the flow of diluting gas, in some embodiments.

Although the outer ring channel may be within the dilution vessel, in some embodiments the outer ring is around the outside envelope of the dilution channel and the radial channels run within radial arms that extend from the inner surface of the envelope, the channels running through the dilution vessel wall to be in fluid communication with the outer ring channel.

Although, the flammable gas diluter is suitable for diluting a variety of flammable gasses, it is particularly suitable for hydrogen. Hydrogen is a very light gas that makes it difficult to pump effectively. It is also prone to accumulate towards the top of any system if flow rate is low. Thus, there are challenges with abating hydrogen and the diluter of embodiments are particular effective at dealing with these challenges. In particular, owing to the reliability of the system with few moving parts, the flow rate is generally maintained at a relatively constant value and the hydrogen will diffuse effectively through the multiple apertures. Furthermore, hydrogen is present in the atmosphere and thus can be vented to the atmosphere when dilute enough. Generally, the flammability limit for hydrogen is a concentration of 4% and thus, below 4% the hydrogen should be safe from ignition. However, to provide a robustly safe system generally a dilution level of 1% at the outlet is the limit that is set.

In some embodiments, said dilution vessel, flammable gas inlet formation and gas directing formations are formed from metal and are earthed.

As noted previously, it may be preferable to remove sources of ignition from the gas flow in the diluter and thus, in some embodiments the flammable gas diluter is formed of a metal that is earthed to reduce any possibility of electrostatic sparks. In some embodiments the metal is stainless steel. In this way, by providing a metallic diluter and having no moving parts within the flammable gas flow and having

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a flow speed above the speed of the flame of the flammable gas in some embodiments, the chances of any ignition are substantially removed.

Although, the inwardly directed gas flow formations and the outwardly directed gas flow formations may have a number of forms, in some embodiments, said inwardly directing gas flow formation comprises a ring-like baffle projecting in from said outer envelope and said outwardly directing gas flow formation comprises a centrally located cone shaped baffle an apex of said cone facing said inlet of said dilution vessel.

These gas flow formations that direct the gas towards the centre and then away from the centre of the gas flow passage through the dilution vessel provide effective mixing of the gasses and thus, effective dilution of the flammable gas within a relatively small volume. This leads to a compact dilution vessel which is convenient to use as a point of use abatement system.

In some embodiments, said plurality of inwardly and outwardly directing gas flow formations are arranged alternatively along a length of said dilution vessel. Thus, an inwardly directing gas flow formation is followed by an outwardly directing gas flow formation.

In some embodiments, said dilution vessel comprises a volume of less than 70 litres, said flammable gas diluter being configured to dilute a flow of flammable gas of up to 1000 SLM (standard litres per minute).

The diluter of embodiments is, due to its design, compact and able to dilute a relatively high flow of flammable gas within a relatively small volume. Thus, a 70 litre diluter, in some cases a 50 litre diluter may dilute a flow of flammable gas of up to 1000 SLM.

In some embodiments, said flammable gas diluter comprises a flammable gas sampler adjacent to said outlet, said flammable gas sampler being in fluid communication with a flammable gas sensor, said flammable gas diluter further comprising control circuitry for inhibiting a flow of flammable gas to said diluter in response to said flammable gas sensor indicating a concentration of flammable gas above a predetermined level.

In order to ensure the safety of the system the concentration of the gas output from the diluter should be below the flammable limit for that gas and thus, in some embodiments there is a sampler close to the output to ensure that this is the case. Were the concentration to rise above the desired flammable concentration limit then hard wired safety control circuitry will inhibit the flow of flammable gas to the diluter. In this regard, where it is a point of use diluter this may involve closing down the tool to which it is attached and which is generating the hydrogen flow.

A second aspect provides a vacuum pumping system for evacuating at least one vacuum chamber in a semiconductor processing tool, said vacuum pumping system comprising: a plurality of vacuum pumps for evacuating said at least one vacuum chamber; and an abatement system for receiving an exhaust from at least one of said at least one vacuum chamber, wherein said abatement system comprises a flammable gas diluter according to a first aspect.

The diluter of the first aspect provides an effective abatement system which can be provided as an integrated system with a vacuum pumping system for evacuating chambers in a semiconductor processing tool that exhausts a flammable exhaust gas. In this way, there is point of use abatement of the flammable gas and there is no requirement to pipe it elsewhere or to burn it in an abatement system with the drawbacks that such burner abatement systems have.

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In some embodiments, said semiconductor processing tool comprises an extreme ultraviolet lithography tool and said flammable gas comprises hydrogen.

Extreme ultraviolet lithography is a technique which uses increasing amounts of hydrogen and thus, an abatement system which provides effective dilution of this hydrogen and can safely dilute it at the point of use without the requirement to pipe it elsewhere or to burn it is a particularly efficient way of evacuating and abating such a system.

In some embodiments, the vacuum pumping system further comprises a housing for housing said plurality of pumps; and a vacuum system air flow generator configured to cause an air flow to flow through said housing to an air flow duct; said air flow duct being in fluid communication with said air flow inlet assembly for supplying said air to said flammable gas diluter.

An air flow is required for dilution of the flammable gas and in some cases there may be a cabinet extract flow, that is a flow of air that is passed over the pumps that are evacuating the system to remove any gasses that might potentially be leaking from the pumps which leakage may be problematic, particularly where they are flammable gasses. Thus, there is already a flow of air in place in many processing systems that pump flammable gasses and this flow of air may be directed directly to the diluter thereby saving in both ducting and air flow generators. This provides an efficient way of re-using the air flow through the housing of the pumping system to dilute the hydrogen that is evacuated from the chamber.

Further particular and preferred aspects are set out in the accompanying independent and dependent claims. Features of the dependent claims may be combined with features of the independent claims as appropriate, and in combinations other than those explicitly set out in the claims.

Where an apparatus feature is described as being operable to provide a function, it will be appreciated that this includes an apparatus feature which provides that function or which is adapted or configured to provide that function.

The Summary is provided to introduce a selection of concepts in a simplified form that are further described in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of the present invention will now be described further, with reference to the accompanying drawings, in which:

FIG. 1 shows a flammable gas diluter according to an embodiment;

FIG. 2 shows a dilution vessel according to an embodiment;

FIG. 3 shows a further view of a flammable gas diluter according to an embodiment;

FIG. 4 schematically shows the flow paths through the dilution vessel of an embodiment;

FIG. 5 shows a vacuum and abatement system according to an embodiment; and

FIG. 6 shows the flow of the gases through a diluter according to an embodiment.

DETAILED DESCRIPTION

Before discussing the embodiments in any more detail, first an overview will be provided.

Dilution is an acceptable option for discharging many flammable gases from a system such as for discharging hydrogen from an EUV tool. Conventionally dilution of a flammable gas has been done using inert gases, however, with increased hydrogen flow, dilution with an inert gas is not acceptable because of the high flow rate required and the added risks and environmental damage that it brings.

Dilution with air provides a cost effective and environmentally friendly alternative.

Basic Theory

Initially the flammable gas from the system such as hydrogen from the EUV contains little or no oxygen and is therefore above the UFL (upper flammable limit) which for hydrogen is 75% hydrogen by volume.

Embodiments seek to dilute to below the lower flammable limit LFL (4% hydrogen by volume). In order to pass from above the UFL where it is safe to below the LFL where it is also safe the mixture must pass through the flammable range (75% to 4% hydrogen by volume). Ignition of the flammable mixture will cause deflagration/detonation. The risk scales with the volume of flammable gas (hydrogen) and the size and lengths of the ducts used to transport the gas.

Hydrogen Characteristics

MIE (minimum ignition energy) for H₂ is 17 μ J

Compared to

Gasoline 800 μ J

Methane 300 μ J

Flame speed for H₂ in air is 2.88 m/s

Compared to

Gasoline ~0.2-0.5 m/s

Methane 0.356 m/s

Provided that the air velocity at the point of H₂ introduction >> flame speed of H₂ (30 m/s) back flash risk is avoided or at least reduced.

Hydrogen gas is highly diffusive and highly buoyant; it rapidly mixes with air.

Risks

It is very easy to ignite a hydrogen mixture.

Once ignited the flame will travel very fast.

In order to mitigate these risks it would be advantageous for dilution with air to be achieved at the earliest opportunity close to the point of use. This reduces the range of locations in which a flammable mixture could be ignited, and reduces the risk of detonation in pipelines and provides a design that can be tested, certified and replicated in any location without needing to analyse the pipe routing and environment every time.

System

Embodiments of the dilution unit seek to dilute the flammable gas to below its lower flammable limit, in some cases to below a quarter of its lower flammable limit and to manage the flammable zone where the flammable gas is between its LFL and UFL so that potential ignition sources are eliminated or at least reduced.

In some embodiments the system utilises the cabinet extract air as the diluent to reduce the need for additional fans for extracting the air from the pumping system cabinets.

FIG. 1 shows a diluter 5 according to an embodiment. Diluter 5 comprises dilution vessel 10 which runs from inlet 11 to outlet 12. Inlet 11 is connected to an air provision portion 14 comprising two air inlets 16 and 17. These are configured to receive air from a primary fan in the case of air inlet 16 and a backup fan in the case of air inlet 17. The air from the air inlet is sent along the pipe in the air provision unit 14 to the air inlet 11 of the dilution unit. The dilution unit 10 has a constricted portion 18 within which a flammable gas inlet arrangement 15 is provided. Thus, the air

when it reaches the constricted portion is accelerated and thus, passes the flammable gas inlet formation at an increased speed and at a speed that is greater than the flame speed of the flammable gas. The dilution unit 10 then expands to a larger diameter and the gasses slow which improves mixing.

The gasses then flow past various gas directing formations to direct them either towards or away from the outer envelope to improve mixing of the flammable gas with the air such that by the time the gasses reach the outlet 12 they are mixed sufficiently that they have a uniform concentration of below the lower flammable limit of the flammable gas. In this regard, the flammable gas entering at inlet 15 is generally at a concentration above the upper flammable limit and it passes through concentrations where it is indeed flammable as it passes through the diluter until it reaches a concentration below the lower flammable limit prior to exit at the outlet 12.

The dilution vessel 10 and in particular, the gas directing formations for encouraging mixing of the flammable gas and the air are shown in greater detail in FIG. 2. In this embodiment, these gas directing formations take the form of cones 30, 34 for directing flow towards the outer walls and baffles 32, 36 for directing flow towards the middle.

The flammable gas enters the dilution vessel at the constricted portion via an inlet arrangement in the form of inlet spider 15. This takes the form of a collar or ring around the outside of the gas dilution vessel 10 into which the hydrogen gas flows. There are radial arms extending through the walls of the dilution vessel from the outer ring and into the gas flow portion. These extend across the cross-section of the constricted portion and apertures on the arms dispense hydrogen into the air flow. These apertures face towards the air inlet 11 of the gas dilution vessel 10.

The gas directing formations 30, 32, 34, 36 are arranged at different longitudinal positions along the length of the dilution vessel 10 and comprise alternatively a cone for directing flow towards the outer edge of the dilution vessel and a baffle for directing it back towards the middle. Thus, at the lower end close to the air and hydrogen inlets there is a cone 30 that diverts the gas mixture towards the outer walls of the vessel and also acts to slow the flow that has been accelerated by constriction 18. Next on the gas flow path is a baffle 32 to direct the air back towards the centre and this is followed by cone 34 and then by baffle 36. In this embodiment there is a sample spider 40 for sampling the gas prior to the exit. This can be directed to a hydrogen sensor to determine that the concentration of the hydrogen exiting the outlet 12 is below the lower flammable limit for hydrogen. The signal from the hydrogen sensor may in turn be sent to control circuitry which in response to determining that the level of hydrogen concentration is above a predetermined limit will generate a control signal for shutting down the hydrogen supply to the diluter, which may involve shutting down the processing in the vacuum chamber from which the hydrogen is received.

FIG. 3 shows the diluter 10 arranged connected to ducts and air flow generators. The air flow generators are in the form of fans 20 and 22 and are attached to air inlets 16 and 17 (see FIG. 1) of the diluter. The fan 20 is the primary fan and fan 22 is the secondary or backup fan. Dilution vessel 10 is shown and one can see the arms from the spider and the apertures on the arms. Although these apertures appear to be facing towards the outlet, in many embodiments they will be facing towards the inlet as gas mixing and flow from these apertures is found to increase with this arrangement. The arms themselves are designed such that they do not

unduly impede the air flow and yet provide a flammable gas flow at different parts of the air flow leading to improved mixing and inhibiting the generation of flumes.

In this embodiment, there is a cabinet extract pipe **64** which receives air that flows around the pumping cabinet comprising the vacuum pumps of the system and which air flow is used to inhibit flammable gasses that may leak from the pumps or their pipes collecting in the cabinet. This air flow is reused in this embodiment as the source of dilution air. This saves on additional fans for pumping this air to the roof and indeed on additional piping.

FIG. **4** schematically shows the flow of gasses through the diluter and how the velocity of the gas flow and the concentration of hydrogen varies along the length of the diluter. The left-hand figure shows the gasses from the air inlet **16** and from the hydrogen inlet **15** and how the air flow is fast and slows at the constriction where the hydrogen is added, whereupon it accelerates again towards exhaust **12**, with some slowing at each of the deflectors **30**, **32**, **34**, **36**. This improves mixing and allows mixings to occur in a relatively small volume.

The right-hand figure shows the molar fraction of hydrogen as the flow progresses through the dilution vessel and mixing occurs. Thus, it goes from a high concentration where it is above the upper flammable limit to a concentration where it is within the flammable limit and then to a concentration where it is below the flammable limit and can be safely exhausted from the vessel. In this embodiment there are 3 cones, **30**, **34**, **38** and after the third cone the molar concentration is about 1% thus, at the required level. As can be seen there is effective mixing of the hydrogen in the primary flow as it leaves uniformly from all the openings of the spider.

FIG. **5** shows a vacuum pumping and abatement system according to an embodiment. This embodiment is for pumping an extreme ultra-violet radiation lithography process. This arrangement comprises pumps in section **50** for pumping an exposure chamber where the wafer is exposed to the laser light and pumps in section **52** for pumping the source chamber where the extreme ultra-violet radiation is generated from lasers and a flow of tin and where hydrogen is used as a shield to protect the various optical components from tin sputtering. The generated extreme ultra-violet light is fed to the exposure chamber via a channel using optical components. Thus, the amount of hydrogen within the source chamber is significantly higher than that within the exposure chamber and it is this hydrogen that is the principal component to be diluted.

In this embodiment, there is a housing **62** which houses the multiple pumps **60** forming the pumping sections **50** and **52**. There is a gas flow through this housing which is the cabinet extract flow which is used to both cool the pumps and to remove any gasses that may leak from them. In this embodiment, there is a duct **64** which channels the cabinet extract gas from the cabinet towards the gas diluter **10** of an embodiment. At the gas diluter **10** there are fans **20** and **22** which feed the air into the gas diluter vessel **10** and an inlet **15** where the hydrogen which is pumped from the foreline of the pumps which evacuate the chambers is input. This input is arranged at the restricted portion of the diluter and comprises the spider. Mixing occurs and the diluted gas is exhausted via exhaust gas flow **66**. In this embodiment fans **20** and **22** are arranged side by side, thereby reducing the height of the diluter.

In this embodiment there are sensors **70** and **72** for sensing both the oxygen and the hydrogen levels within the different flows. There may also be a hydrogen sensor close

to the output of the diluter **10** and these sensors can be used with control circuitry to inhibit the process if it is determined that the concentration levels of hydrogen or oxygen are such that there may be flammability problems with the gas being exhausted.

FIG. **6** schematically shows the different zones within the dilution system. There is the initial zone where extracted air is input through primary and backup centrifugal fans. The extracted air has very low levels of hydrogen within it and therefore is below the low flammable limit. Gas is received via a conduit from the scanner or exposure chamber **100** and the source chamber **120** with respective compositions of 132 slm (standard litres per minute) N_2 and 12 slm H_2 from the scanner chamber and 600 slm H_2 from the source chamber. Within the conduit the concentration of the hydrogen is above the upper flammable limit, while the oxygen levels are below the limiting oxygen concentration. The hydrogen is input via an input spider **15** and there is a flame arrester cone **75** to impede ignition of this gas flow when it meets the air. The gas flow flowing through the zone here (zone 0) is above the lower flammable limit and below the upper flammable limit and above the limiting oxygen concentration so that ignition is possible. In order to avoid ignition or at least impede it there are no moving parts within this portion of the diluter, the material of the diluter is formed of an earthed metal and the flow rate is kept above the flame speed of the flammable mixture.

As the flammable mixture flows through the dilution vessel mixing occurs due to the baffles and cones until at zone two the concentration of the flammable material has dropped to below a quarter of the lower flammable limit and the gas can now be safely vented to air. There are flammable gas sensors **72** at both the outlet and further into the dilution vessel to check that the gas mixture is substantially below the lower flammable limit.

There are primary and backup fans **20**, **22** which in this embodiment are centrifugal fans with speed control able to provide a flow speed of up to 4000 m³/hr. This provides a velocity of the input air of about 14 m/s which when accelerated at the constriction **18** increases to above 30 m/s which is about 10 times the flame speed of hydrogen.

The distribution of the hydrogen into the air stream also prevents flashback and the cones and baffles encourage mixing of the hydrogen into the air stream such that the concentration of hydrogen is <1% v/v by the time it reaches the last baffle plate.

In summary, embodiments provide a system that uses the cabinet extraction air as the dilution air, aggregates air from parallel paths by using a multiple aperture inlet spider, provides the motive force from an integral blower, and in this way provides a defined flammable zone. Critical parameters such as flammable gas concentration, air flow and temperature are monitored and interlocked to control signals, such that control of the system and/or shutdown of the system can be provided in response to these signals.

In some embodiments there is a second fan available to backup the primary fan. The fans can be maintained and serviced during tool operation as no H_2 flows through the fans.

Embodiments of the diluter provide a system which has few moving parts and uses redundancy and diversity in much of its monitoring and control system such that it is reliable and insensitive to common mode failures. In particular, there is a back up fan, the air flow comes from the cabinet extract for which flow balancing is provided across pump modules.

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In some embodiments, one 4 kW inverter driven fan is sufficient to provide both cabinet extract and dilution. The second fan is in standby and preventative maintenance monitoring of the operational fan is used to detect when the standby fan needs to be brought online. In the event that any degradation in primary fan performance is detected, the secondary fan will start up and the primary fan will be isolated and stopped. The bearings and the motor can be maintained in-situ for each fan. The fan inlet and outlet are independently isolated.

In some embodiments, it takes 5 seconds for the secondary fan to run up to full speed. This is significantly more cost and power efficient than with a fueled burner where both primary and backup units run.

Furthermore, as the dilution unit is a point of use unit, there is no need to extract from systems that are not running or that are running only Scanner hydrogen flows.

In the event of a NOK signal the diluter will continue to operate and dilute.

In the event that there is a system shutdown, the pumping system removes the OK signal that shuts off the H₂ supply to the tool.

In the event of an unexpected stop, the residual H₂ will equalise within the pumping system and as the system stops some of the gas will remain in the system and some will enter the Diluter and diffuse with any air and enter the facilities exhaust. The diluter will restart followed by the pumping system and purge in the same way it does currently after an unexpected stop.

Although illustrative embodiments of the invention have been disclosed in detail herein, with reference to the accompanying drawings, it is understood that the invention is not limited to the precise embodiment and that various changes and modifications can be effected therein by one skilled in the art without departing from the scope of the invention as defined by the appended claims and their equivalents.

Although elements have been shown or described as separate embodiments above, portions of each embodiment may be combined with all or part of other embodiments described above.

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are described as example forms of implementing the claims.

The invention claimed is:

1. A flammable gas diluter for diluting a flow of flammable gas to a concentration below the flammability limit of said flammable gas, said diluter comprising:

a dilution vessel comprising an outer envelope defining a longitudinal flow passage from an inlet to an outlet; at least one air inlet assembly for directing a flow of air into said inlet of said dilution vessel;

a flammable gas inlet arrangement comprising a plurality of apertures at least some being arranged at different distances from said outer envelope of said dilution vessel across a cross section of said dilution vessel towards said inlet end of said dilution vessel;

a plurality of gas flow directing formations arranged between said flammable gas inlet arrangement and said outlet, each being at a different position along a length of said dilution vessel; wherein

at least one of said plurality of gas flow formations is an inwardly directing gas flow formation for directing gas flow away from said outer envelope and at least one of

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said gas flow formations is an outwardly directing gas flow formation for directing gas flow towards said outer envelope; and

wherein said dilution vessel comprises a constricted portion, said flammable gas inlet arrangement being located within said constricted portion such that said air is accelerated prior to passing said flammable gas inlet.

2. The flammable gas diluter according to claim 1, comprising at least one gas flow generator for pumping a flow of air into said air inlet assembly, said at least one gas flow generator being located upstream of said flammable gas inlet arrangement.

3. The flammable gas diluter according to claim 2, comprising two gas flow generators configured to operate as operational and backup gas flow generators.

4. The flammable gas diluter according to claim 2, wherein said at least one gas flow generator and dilution vessel are configured such that a speed of flow of said air at said flammable gas inlet arrangement is greater than a flame speed of flammable gas.

5. The flammable gas diluter according to claim 1, wherein said flammable gas inlet arrangement is configured such that said apertures are facing away from said gas outlet.

6. The flammable gas diluter according to claim 5, wherein said apertures of said flammable gas inlet arrangement are between 2 and 5 mm in diameter.

7. The flammable gas diluter according to claim 1, wherein said flammable gas inlet arrangement comprises an outer ring channel and radial channels running from said outer ring channel towards a centre of said ring, said radial channels comprising said apertures.

8. The flammable gas diluter according to claim 1, wherein said flammable gas comprises hydrogen.

9. The flammable gas diluter according to claim 1, wherein said dilution vessel, flammable gas inlet arrangement and gas directing formations are formed from metal and are earthed.

10. The flammable gas diluter according to claim 1, wherein said inwardly directing gas flow formation comprises a ring-like baffle projecting in from said outer envelope and said outwardly directing gas flow formation comprising a centrally located cone shaped baffle an apex of said cone facing said inlet of said dilution vessel.

11. The flammable gas diluter according to claim 1, said dilution vessel comprising a volume of less than 70 litres, said flammable gas diluter being configured to dilute a flow of flammable gas of up to 1,000 standard litres per minute.

12. The flammable gas diluter according to claim 1, said flammable gas diluter comprising a flammable gas sampler adjacent to said outlet, said flammable gas sampler being in fluid communication with a flammable gas sensor, said flammable gas diluter further comprising control circuitry for inhibiting a flow of flammable gas to said diluter in response to said flammable gas sensor indicating a concentration of flammable gas above a predetermined level.

13. A vacuum pumping system for evacuating at least one vacuum chamber in a semiconductor processing tool, said vacuum pumping system comprising:

a plurality of vacuum pumps for evacuating said at least one vacuum chamber; and

an abatement system for receiving an exhaust from at least one of said at least one vacuum chamber, wherein said abatement system comprises a flammable gas diluter according to claim 1.

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14. The vacuum pumping system according to claim **13**, wherein said semiconductor processing tool comprises an extreme ultraviolet lithography tool and said flammable gas comprises hydrogen.

15. The vacuum pumping system according to claim **13**,
further comprising a housing for housing said plurality of
pumps; and

a vacuum system air flow generator configured to cause
an air flow to flow through said housing to an air flow
duct;

said air flow duct being in fluid communication with said
at least one air inlet assembly for supplying said air to
said flammable gas diluter.

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